2026 IEEE Mass Storage Roadmap

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The 2026 IEEE Mass Data Storage Roadmap has 156 pages, with 12 tables and 84 figures analyzing the current state and projecting the future of solid-state non-volatile memory, magnetic data storage including hard disk drives as well as magnetic tape, optical storage and DNA storage developments. The enormous growth of digital storage demand to support AI training and inference is driving demand all current commercial digital storage technologies throughout the digital storage tiers.

***Index Terms*—IEEE, roadmap, mass storage, Solid state drive, SSD, non-volatile memory, hard disk drive, HDD, magnetic tape, optical storage, DNA storage.**

I. Introduction

In 2025 and early 2026 experts on various digital storage and non-volatile memory technologies worked together to update the 2023 International Roadmap for Devices and Systems on Mass Data Storage [1]. The roadmap report including information on the future of solid start storage and non-volatile memory, hard disk drives, magnetic tape, optical storage and DNA storage. This digest summarizes some of the results from this roadmap update.

II. General Obervations

AI demands data and data lives on digital storage and memory. In 2025 and going into 2026 supply has been much less than demand and this has resulted in increasing prices for all shipping digital storage technologies. Many of the storage and memory companies have reported having their planned production capacity sold out through 2026 and into 2027 with some long-term contracts out to 2028. Supply is gradually expanding, but storage and memory companies are being cautious after the downturn in demand in 2022-2023 and also because, particularly for solid state storage using NAND flash, new production may take 18 months or more to come online.

These factors make developments in digital storage and non-volatile memory an important consideration for the future of computing. The report projected the future of the major commercial storage and non-volatile memory technologies as well as new optical and DNA technologies geared for archiving and preservation of data.

III. Solid State Non-volatile Memory

In the semiconductor memory market, the size of the NAND flash memory market is second only to DRAM. In 2025, the size of the NAND flash market is estimated to be $68B dollars [2]. AI demand in 2025 and 2026 is driving revenue and prices higher as demand exceeds current NAND flash supply.

NAND flash has replaced other types of memory in consumer, industrial and enterprise applications. NAND flash was able to do this by decreasing its cost and providing higher performance than other competing storage technologies. There are challenges to future advances, but the roadmap indicates that NAND flash will continue to be dominant at least into the 2030's.

Non-volatile memory technologies such as MRAM, ReRAM, PCM and FRAM have some standalone chip applications but their highest volume manufacturing is as embedded memory in devices from the major chip foundries. There they are replacing NOR flash and supplementing slower SRAM [3].

Advances in these technologies and higher embedded manufacturing volume could see one or more of these memories achieving the manufacturing volume needed to reduce their costs of manufacturing in order to compete successfully against DRAM and even NAND flash as scaling issues slow these established memory technology advances.

IV. magnetic Recording Technology

A. Hard Disk Drives

Of current mass data storage technologies, in terms of storage capacity shipped, hard disk drives (HDD) are by far the largest single component of the mass data storage industry. The growing demand for low-cost HDD storage for data centers has stabilized the annual unit shipments. In addition, the introduction of new recording technologies, particularly HAMR, has resulted in significant storage capacity cumulative annual growth and the total shipped HDD storage capacity has grown considerably.

Seagate and Western Digital are shipping 30+TB HDDs with over 100TB HDDs expected before 2030. **Table 1 d**isplays the current roadmap outlook of key attribute needs for Hard Disk Drive (HDD) storage technology.

A. Magnetic Tape

Magnetic tape technology is particularly well suited to help meet AI demand due to its very low total cost of ownership and its efficient data center footprint that is more than 7.74 PB/ft^2. Part of the TCO advantage of tape arises from its very low power consumption that also contributes to its small CO2e footprint. The natural physical airgap of tape solutions also provides an additional level of protection against accidental data deletion and cyber security threats.

Moreover, recent tape areal density demonstrations provide confidence that tape has the potential to continue scaling areal density for multiple future generations with cartridge capacities expected to reach hundreds of TB per cartridge within the next decade. Table 2 summarizes a set of major tape industry metrics and their expected evolution over the next 15 years.

TABLE I
HARD DISK DRIVE ROADMAP

	Unit	2022	2025	2028	2031	2034	2037	2040
Industry Metrics								
Form Factor (dominant form factor is bold)	inches	3.5, 2.5	3.5, 2.5	3.5, 2.5	3.5, 2.5	3.5	3.5	3.5
Capacity	TB	0.5-22	0.5-36	2-60	7-80	8-100	10-150	12-200
Market Size	units (M)	172	124	136	148	160	180	200
Cost/TB (avg.)	$/TB	13.6	6.91	3.46	2.60	2.00	<2.00	<1.80
Design/Performance								
Areal Density	Tb/in^2	>1.0	>2.0	>4.0	>6.0	>8.0	>10.0	>12.0
Rotational Latency	ms	2-12	2-12	2-12	2-12	3-12	3-12	3-12
Seek Time*	ms	3-5	3-5	3-5	2-5	1.5-5	1-4	0.5-3.5
RPM		4.2-10K	4.2-10K	4.2-7.2K	4.5-7.2K	4.5-7.2K	4.5-7.2K	4.5-7.2K
Data rate	MB/sec	140-600	140-600	180-1,200	200-1,200	210-1,800	220-1,800	230-1,800
Power	watts	1.9-14	1.9-14	3-14	3-14	3-14	3-14	3-14
Key Component Requirements								
Read Head	type	TMR	TMR	TMR	TMR	TMR / CPP-GMR	CPP-GMR / LSV	CPP-GMR / LSV
Slider	type & size (% of micro, 3.86 mm^3)	5%	5%	5%	5%	<5%	<5%	<5%
Disk	type	AlMg, Glass	AlMg, Glass,	AlMg, Glass,	Glass, New Substrate,	Glass, New Substrate,	Glass, New Substrate,	Glass, New Substrate,
Disk Static Coercivity	Oe	5,000-6,000 HAMR ≥30,000	5,000-6,000 HAMR ≥30,000	5,000-6,500 HAMR ≥30,000	5,000-20,000 HAMR ≥30,000	6,000-40,000	20,000-50,000	20,000-50,000
Magnetic Recording Technology		Perpendicular TDMR, ePMR, HAMR, MAMR, SMR	Perpendicular TDMR, HAMR, SMR	Perpendicular TDMR, HAMR, SMR	Perpendicular TDMR, HAMR, SMR	Perpendicular TDMR, HAMR, SMR	Perpendicular TDMR, HAMR, SMR, Heated Dot Recording	Perpendicular TDMR, HAMR, SMR, Heated Dot Recording
	Unit	**2022**	**2025**	**2028**	**2031**	**2034**	**2037**	**2040**
Electronics/Channel	type	LDPC Iterative GPR (Turbo), Pattern Dependent Noise Predictive GPR	LDPC Iterative GPR (Turbo)	LDPC Iterative GPR (Turbo), TDMR	LDPC Iterative GPR (Turbo), TDMR	Soft ECC, TDMR	Soft ECC, TDMR	Soft ECC, TDMR
Channel Bandwidth	MHz	500-2,000	500-2,000	500-2,000	500-2,500	>2,500	<3,000	~3000
SNR	dB	<20	<20	<20	<20	<18	<18	<17
Actuator	type	Conventional/Micro, +DSA	Conventional/Micro, +DSA	Micro, + DSA	Micro, + DSA	Micro, + DSA	Micro, + DSA	Micro, + DSA
Spindle	type	Fluid	Fluid	Fluid	Fluid	Fluid	Fluid	Fluid

*Seek time is one third full stroke seek time and does not include micro-actuator local track or rotational latency.

TABLE 2
MAGNETIC TAPE ROADMAP

	Jnit	2022	2025	2028	2031	2034	2037	2040
Native Capacity	TB	20	40	70	120	210	365	640
Max Data Rate (native)	IB/s	400	400	600	900	1350	2000	3000
Areal Density	ib/in^2	11.7	23.2	32.1	53.3	88.4	146.7	243.5
Tape Speed (for data)	n/s	5.8	5.8	3.9	5.4	7.3	5.0	6.8
Volumetric Density	TB/in^3	1.5	3.0	5.3	9.2	16.1	28.0	48.8
Form Factor*	a	HH, FH	HH, FH	HH*, FH	HH*, FH	HH*, FH	HH*, FH	HH*, FH
Number of channels	a	32	32	64	64	64	128	128
Magnetic Film	a	BaFe	BaFe SrFe	BaFe SrFe ε-FeO	SrFe ε-FeO	SrFe ε-FeO	SrFe ε-FeO	SrFe ε-FeO Sputtered
Recording Tech.		Perp.	Perp.	Enhanced Perp.	Enhanced Perp.	Enhanced Perp.	Enhanced Perp.	P+SUL
Tape Thickness	um	5.20	5.20	4.99	4.79	4.60	4.42	4.24
Substrate Material	ype	PEN Aramid Spaltan	PEN Aramid Spaltan	PEN Aramid Spaltan	PEN Aramid Adv. Sub.	PEN Aramid Adv. Sub.	PEN Aramid Adv. Sub.	PEN Aramid Adv. Sub.

es: Volumetric Density: Capacity divided by volume of an LTO cartridge (4 x 4.1 x 0.8 = 13.12 in^3)
Form Factor: HH = half height = 5.25" half-height internal form factor
FH = half height = 5.25" full-height internal form factor
*Note, due to power density and space constraints HH drives will likely remain in a channel format with lower data rate when FH drives transition to a 64-channel forma
Recording Technology: Perp. = perpendicularly oriented media,
Enhanced Perp. = perpendicular media with improved orientation
P+SUL = perpendicular media with a soft underlayer
Substrate Material: PEN = polyethylene napthalate,
Aramid = Aromatic Polyamid
Adv. Sub. = Advanced Substrate

V. OPTICAL DATA STORAGE

Optical storage media has undergone a shift from its traditional role in consumer media distribution to a focus on enterprise and institutional archival storage. To enhance capacity and long-term data preservation a number of optical storage start-ups are currently active. Some of these technologies are based upon conventional looking optical disc and others are focusing on rectangular media that could fit into cartridges in a magnetic tape library. Table 3 shows a projected optical storage roadmap for existing and new optical storage technologies.

TABLE 3
OPTICAL STORAGE ROADMAP

Technology	Metrics	Unit	2024	2025	2026	2030	2035	2040
BDXL	Capacity	TB	0.2	0.2				
	Media Cost	$/TB	40	40				
	Data Transfer Rate (Max)	MB/sec	27	36				
	Layer Count (Double Side)	Layers	6	6[xix]				
Archival Disk No Further Roadmap Announced	Capacity	TB	0.5	0.5				
	Media Cost	$/TB	33	30				
	Data Transfer Rate (Max)	MB/sec	375	375				
	Layer Count (Double Side)	Layers	6	6				
					2027-2028			
Folio Photonic Disc[xx]	Capacity	TB			0.5 - 1.0	1 - 2	4 - 8	10 - 30
	Data Transfer Rate (Max)[xxi]	MB/sec			40	320	600	600
	Layer Count (Double Side)	Layers			16	32	64	128
SPhotonix	Capacity	TB			20	100	250	500
	Data Transfer Rate (Max)	MB/sec			100	300	500	500
	Layer Count (Double Side)	Layers			100	200	500	1000
Cerabyte (Rack Level Solution)	Capacity per Cartridge	TB			0.5	20-40	150	500
	Data Transfer Rate (Max)	MB/sec			100	250	1000-2000	2000
	Layer Count (Double Side)	Layers			1	1	1	1

VI. DNA DATA STORAGE

DNA as a storage medium is compelling due to its extreme density, zero power at rest, durability, and format immutability (which, together with durability, eliminates the need for technology migration). While it is still uncertain when or whether a commercially viable DNA data storage ecosystem will emerge, with the continuing advances in medical/scientific DNA technology and applications, plus academic research and commercial biotech both targeted at DNA data storage over the past decade and a half, the basic foundations for synthetic DNA as a data storage medium have been demonstrated on scalable technology platforms.

REFERENCES

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